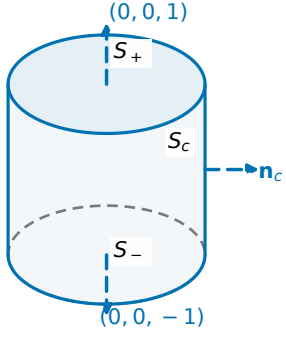
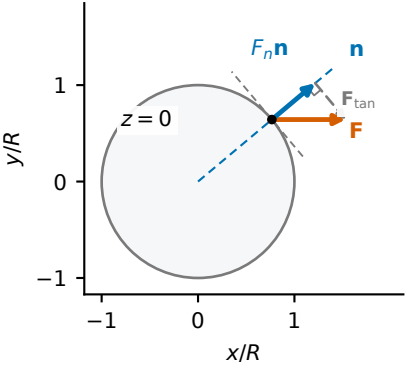
A



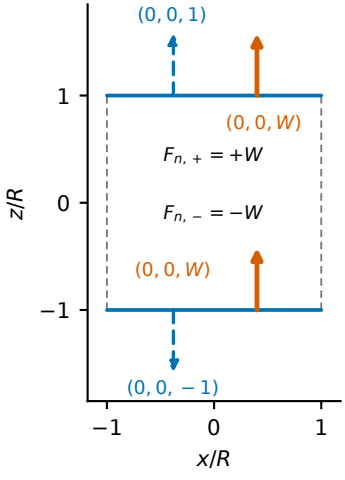
$\mathbf{n}_c = (\cos\theta, \sin\theta, 0)$

B



$$F_n = \frac{R\cos^2\theta}{R^2 + z^2} \geq 0$$

C



$I_\pm = \pm\pi R^2 \ln 2$